



Smart Crop Advisory System (**Krishi Sahayak**)

ML-driven fertilizer optimization and intelligent crop management through advanced data science and deep learning models

Minor Project

Presented By: Abhay Maurya | Amit Yadav | Varun Rana | Shanu Ahmed | Samir Ahmad

Mentor: Mrs. Sneh Lata Singh

Department of CSE: Dr. APJ Abdul Kalam Institute of Technology, Tanakpur



Krishi Sahayak



Crop Recommendation Model

Our machine learning model predicts optimal crops using Using soil report data. Trained on 30,530 samples across 8 features including nitrogen, phosphorus, potassium levels, temperature, humidity, pH, and rainfall patterns.

93.71%

Final Test Accuracy

30K

Training Samples

8

Key Features

Random Forest

Best accuracy: 90.41%

XGBoost

Best accuracy: 89.97%

K-Nearest Neighbors

Best accuracy: 87.60%



Fertilizer Optimization Model

Precision dosage calculation using ensemble machine learning combines regression and classification approaches to deliver optimal fertilizer recommendations based on environmental and soil parameters.

95.75..

Test Accuracy

5,410

Training
Samples

11

Input Features

Model Performance Comparison

- **XGBoost:** 93.60% (best single model)
- **Random Forest:** 93.53% accuracy
- **KNN:** 83.41% baseline

Key Input Parameters

- Temperature & moisture levels
- Rainfall patterns
- Soil pH, NPK composition
- Carbon content & crop type

Plant Disease Detection

Deep learning-powered disease identification analyzes uploaded crop images using Convolutional Neural Networks built on MobileNetV2 architecture for rapid, accurate diagnosis across 34 disease classes.



MobileNetV2 CNN

Optimized architecture for mobile deployment with transfer learning



34 Disease Classes

Corn, pepper, potato, rice, sugarcane, tomato, and wheat diseases



32,640 Training Images

Comprehensive dataset ensuring robust model generalization

Technology Stack

- **Deep Learning:** TensorFlow & Keras frameworks
- **Image Processing:** Pillow (PIL) and NumPy
- **Architecture:** MobileNetV2 with custom classification head
- **Preprocessing:** Automated image augmentation and normalization

📄 **Supported Crops:** Corn, bell pepper, potato, rice, sugarcane, tomato, and wheat with disease-specific detection including bacterial spots, blights, rusts, and viral infections

Intelligent Support Systems



Multilingual AI Chatbot

NLP-powered conversational assistant integrated with translation APIs enables farmers to receive guidance in their native language. Text-to-speech capabilities provide voice support for accessibility and ease of use in field conditions.



WhatsApp Alert System

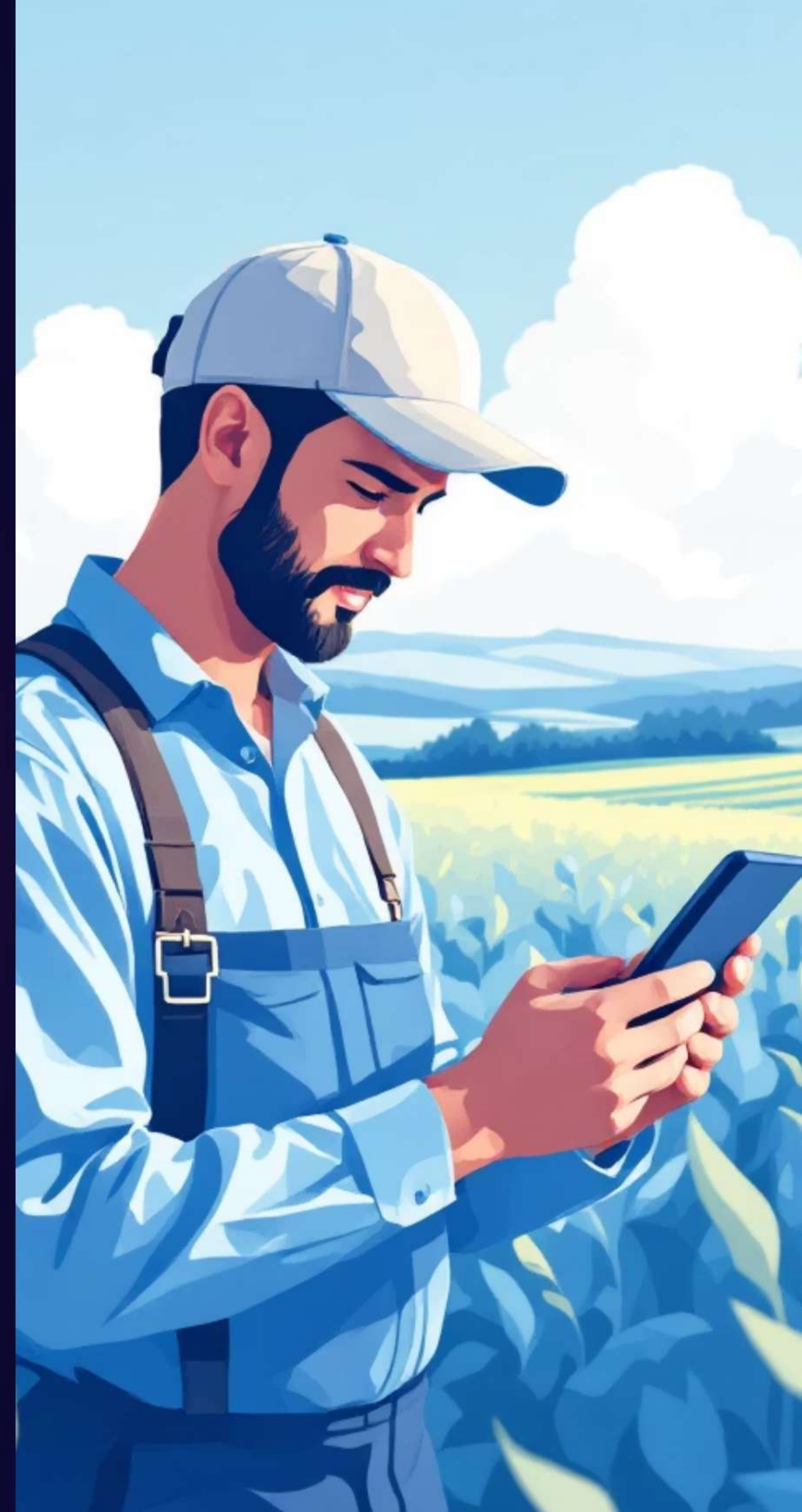
Spring Boot backend triggers automated, real-time notifications for critical events including disease outbreak warnings and severe weather alerts. Proactive messaging helps farmers take preventive action before crop damage occurs.

Chatbot Features

- Multi-language support via translation APIs
- Voice interaction with text-to-speech

Alert Capabilities

- Disease outbreak notifications
- Severe weather warnings
- Personalized crop recommendations



Knowledge Sharing Platform



Dynamic Blog Implementation

Interactive content management system enables agricultural experts and farmers to share insights, best practices, and success stories through a collaborative platform.



Spring Boot Backend

Robust server-side architecture handling business logic and data persistence



JavaScript Frontend

AJAX-powered real-time updates without page refresh for seamless user experience



MySQL Database

Reliable data storage with full CRUD operation support

Technology Stack Overview

This full-stack web application is built on a robust foundation of enterprise-grade technologies, carefully selected to ensure scalability, security, and maintainability. Our stack leverages industry-standard frameworks and tools that provide both development efficiency and production reliability.



Backend Framework

Spring Boot powers the application's core, providing rapid development capabilities and production-ready features out of the box.

Spring Data JPA simplifies database operations with elegant object-relational mapping and reduces boilerplate code.

Spring Security delivers comprehensive authentication and authorization, protecting sensitive data and user sessions.



Frontend Technologies

HTML5 & CSS3 create semantic, responsive layouts that work seamlessly across devices and screen sizes.

JavaScript enables dynamic, interactive user experiences with modern ECMAScript features.

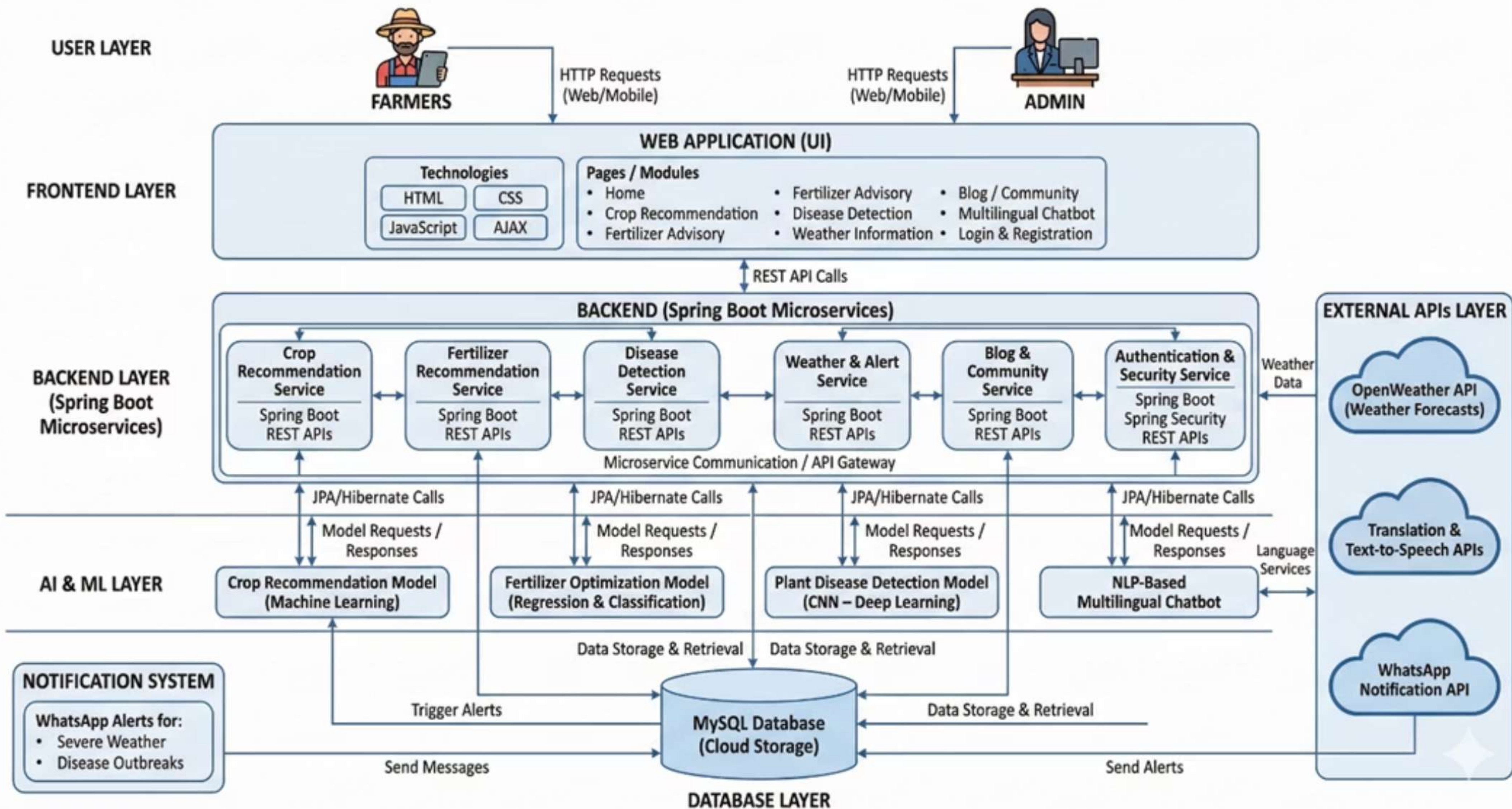
AJAX facilitates asynchronous data exchange, providing smooth, real-time updates without page reloads.



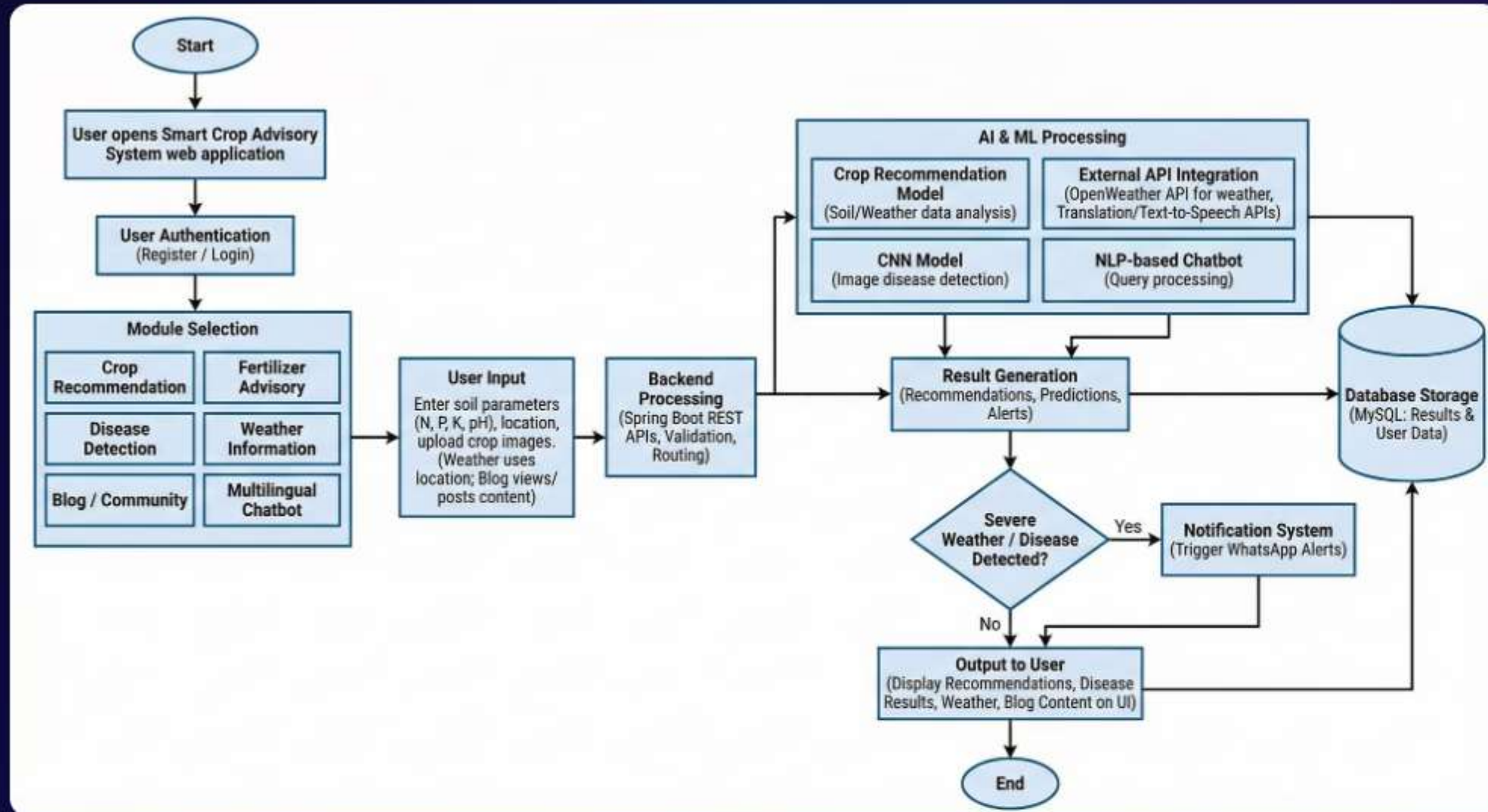
Data Layer

MySQL serves as our reliable relational database management system, offering ACID compliance and excellent performance.

Smart Crop Advisory System (SCAS) - Microservices-Based Architecture



System Architecture Diagram for Smart Crop Advisory System (SCAS).



Workflow Diagram for Smart Crop Advisory System (SCAS).



ER Diagram for Smart Crop Advisory System (SCAS).

Thank You

